WORKSHEET

Chapter 2: Working with formulas

1 The braking distance, d m, of a bicycle travelling at speed V m/s is:

$$d = \frac{V(V+1)}{2}$$

Find the braking distance at a speed of 8.5 m/s.

2 For the formula $A = LB$, find the value of:

a A when L is 12.5 and $B = 9$

b L when $A = 91$ and $B = 6.5$.

3 Given the formula $E = mc^2$, find E if $m = 4.7 \times 10^{-20}$ and $c = 3 \times 10^8$.

4 From a height of h m, a person can see a distance of d km to the horizon, where:

$$d = 8\sqrt{\frac{h}{5}}$$

What distance (to the nearest kilometre) can be seen from the top of Centrepoint Tower, 304 m high?

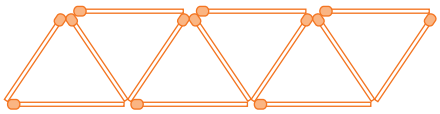
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5 The formula for converting feet to metres is $M = 0.3048F$. Convert, correct to two decimal places:

a 6 ft to m

b 10 m to ft.

6 The number of matchsticks, m, used to make this pattern of triangles is $m = 2t + 1$, where t is the number of triangles.



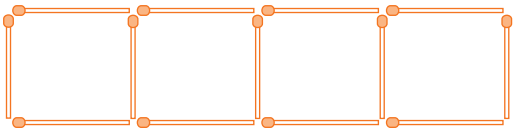
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How many:

a matches are needed to make 65 triangles?

b triangles can be made with 55 matches?

7 **a** Find a formula for calculating the number of matchsticks, m, needed to make a pattern of s squares.



...

b Use your formula to find the number of matchsticks needed to make 15 squares.

8 The area of a trapezium is $A = \frac{1}{2}(a + b)h$.

Find the height of a trapezium that has an area of 52.5 cm^2 and parallel sides of lengths 5 cm and 9 cm.

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9 If $S = \frac{n}{2}(a + l)$, find n when $S = 14\,928$, $a = 25$ and $l = 286$.

10 The volume of a cylinder is $V = \pi r^2 h$. Find, correct to three significant figures:

a the volume of a cylinder with radius 6 m and height 10.5 m.

b the height of a cylinder with volume 12 211.68 cm^3 and radius 8 cm.

11 The formula for converting kilometres to miles is $m = \frac{5k}{8}$

Use it to convert:

a 25 km to miles

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b 8 miles to km.

12 Find the radius of a circle that has a circumference of 43.98 cm. Give your answer correct to two decimal places.

13 Find the height of the triangle with an area of 84 m^2 if its base is 14 m long.

14 The mean of two numbers x and y is $\frac{x + y}{2}$

The mean of 78 and another number is 63. What is the other number?

15 The formula for converting Fahrenheit (°F) temperatures to Celsius (°C) is:

$C = \frac{5}{9}(F - 32)$

a Boiling point is 212 °F What is this temperature in Celsius?

b Convert a temperature of 200 °C to °F

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16 The time T it takes a swing to rock back and forth is $T = 2\pi\sqrt{\frac{l}{g}}$ where l is the length of the swing and g is the gravitational acceleration. Find T to two decimal places if $l = 2.6$ and $g = 9.8$.

17 The angle sum (in degrees) of a polygon with n sides *is* $A = 180(n - 2)$. Find the number of sides of the polygon that has an angle sum of 1440°.

18 The cost of hiring a chauffeured limousine is \$185 plus \$3.70 for each kilometre travelled.

a Calculate the cost of hiring a limousine to travel 22 km.

b Construct a formula that calculates the cost, \$$C$, of hiring a limousine to travel d km.

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Answers

1 40.375 m
2 **a** 112.5
b 14
3 4.23×10^{-3}
4 62 km
5 **a** 1.83 m
b 32.81 ft
6 **a** 131 matches
b 27 triangles
7 **a** $m = 3s + 1$
b 46 matchsticks
8 7.5 cm
9 96
10 **a** 1190 m^3
b 60.7 cm
11 **a** 15.625 miles
b 12.8 km
12 7.00 cm
13 12 m
14 48
15 **a** 100 °C
b 392 °F
16 3.24 seconds
17 10 sides
18 **a** $266.40
b $c = 185 + 3.7d$

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